Plan 08 v0: Learned Memory Wrapper for Long-Context RCA

Progress / Fact Record

Mingjia Shi

RCA Foundation Model / Long-Context Memory

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Reporting period: June 2026, Week 1

Date range: Mon 2026-06-01 to Sun 2026-06-07

Goal: concise progress record; detailed facts live in linked GitHub notes.

This Week: Main Message

High-level direction

Controlled model adaptation: improve RCA / long-context capability while preserving the frozen base model's general ability.

- **Plan 08 v1**: learned soft-prompt memory is now framed as a lossy-compression characterization, not a universal-memory claim.
- **Plan 08 v1.5**: next step is a do-no-harm gate that opens the wrapper only when compression is likely safe.
- **Plan 08 v2**: next version is cross-session latent memory with read/write tokens under a frozen base.
- **Related direction**: Janus tests whether long-context sites and forgetting-sensitive sites are coupled.

Problem Statement: Why This Matters

- RCA and debugging inputs are long: logs, traces, topology, prior observations, and code context often exceed cheap prompting.
- Direct SFT can teach RCA behavior, but risks weakening general instruction-following, reasoning, or retrieval behavior.
- The safer target is a **frozen-base adaptation module**: learn context compression / memory without overwriting the base model.
- This week sharpened the decision rule: use memory when it helps; close it when it would hurt.

Plan 08 v1: Main Result Cell

Benchmark	Wrapper	Gist	Best reference
QuALITY	0.193 ± 0.032	0.180 ± 0.044	no-context 0.141
MuSR-mm	0.493 ± 0.008	0.501 ± 0.013	full-context 0.551
RULER-NIAH	0.000 ± 0.000	0.000 ± 0.000	full-context 0.995

- **Conclusion**: wrapper works as a narrow lossy compressor.
- Helps when gist-level information is sufficient; fails when exact detail preservation is required.
- Full facts: v1 results summary.

Plan 08 v1.5: Signal Grid Cell

Signal family	Interpretation	Direction	Use in gate
Confidence / seq logprob	wrapper likely correct	positive	open
Wrapper-to-base KL / JS / TV	wrapper fights base	inverse	close
Mid/late-layer drift	late reasoning distorted	inverse	close

- **Design move**: gate on "is compression safe?" rather than forcing memory onto every input.
- Main signal-grid deliverables: grid folder, help ranking, non-obvious ranking.
- Compiled note: v1.5 PDF.

Plan 08 v2: Next-Version Setting

- **Question**: can latent memory be written during generation and reused across sessions?
- **Base**: still frozen; adaptation remains wrapper-only.
- **New API**: add ‘update_from_generation‘, ‘serialize‘, and ‘load‘ to the v1 wrapper protocol.
- **Differentiation**: cross-session persistence + frozen base + v1 bit-capacity diagnostic.
- Links: v2 plan, v2 related work.

Deliverables: Where To Read More

Item	Link
Plan 08 hub	README / deliverables hub
v1 facts	main results summary
v1.5 report	compiled PDF
v1.5 signal grid	CSV + figures
v2 setting	v2 plan
Idea tables	ideas index / plans index

Next Steps

1. Implement the first do-no-harm gated wrapper using confidence and divergence features.
2. Evaluate whether gating preserves gains on compressible inputs while closing on unsafe exact-retrieval / reasoning-transfer cases.
3. Keep v1 as the characterization paper line; keep v2 as the cross-session latent-memory next version.
4. Continue Janus causal testing in parallel: protect long-context-coupled sites during SFT and measure forgetting / long-context retention.